

Course: Networked Control

Computer session 03: Control under rate limitations

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Outline

- Control under quantization and selection functions
- Available program
- Exercises

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Control under quantization and selection functions

Consider the continuous-time scalar system

$$\dot{x} = ax + bu$$

where, for simplicity, we assume that $b > 0$.

The dynamics of the system under control is

$$x_{k+1} = fx_k + gu_k$$

where, if $a > 0$ (open-loop unstable system) the control law defined in case we operate at the rate limits is

$$u_k = \begin{cases} u_{\max} & x_k \leq \frac{b}{a} \left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\max} - i \frac{u_{\max} - u_{\min}}{N-1} & \frac{b}{a} \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) < x_k \leq \frac{b}{a} \left(-u_{\max} + (i+1) \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\min} & x_k > \frac{b}{a} \left(-u_{\min} + \frac{u_{\min} - u_{\max}}{N} \right) \end{cases}$$

→ The set $[-\frac{b}{a}u_{\max}, -\frac{b}{a}u_{\min}]$ is invariant.

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Available program: dataratelimit.m

In the WeBeep portal, you can download the function “dataratelimit.m” that, given a system and its characteristics, computes the “standard” control law – defined under strict rate limitations (or its “variations” in case $a \leq 0$) – and applies it to the system.

```
>> help dataratelimit
Control under data rate limitation FOR SCALAR SYSTEMS
dx/dt=ax+bu
It applies the control law defined in case we operate at the
rate limits.
Inputs:
- a: system matrix (scalar)
- b: input matrix (scalar)
- umin: minimum value of u
- umax: maximum value of u
- NB: n. of bits/packet
- h: sampling time
- init_point: initial condition for the simulation
- k_end: n. of simulation samples
```

Available program: dataratelimit.m

If $a > 0$ the applied control law is

$$u_k = \begin{cases} u_{\max} & x_k \leq \frac{b}{a} \left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\max} - i \frac{u_{\max} - u_{\min}}{N - 1}, & \frac{b}{a} \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) < x_k \leq \frac{b}{a} \left(-u_{\max} + (i + 1) \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\min} & x_k > \frac{b}{a} \left(-u_{\min} + \frac{u_{\min} - u_{\max}}{N} \right) \end{cases}$$

If $a < 0$ the applied control law is

$$u_k = \begin{cases} u_{\max} & x_k \leq \frac{b}{|a|} \left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\max} - i \frac{u_{\max} - u_{\min}}{N - 1}, & \frac{b}{|a|} \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) < x_k \leq \frac{b}{|a|} \left(-u_{\max} + (i + 1) \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\min} & x_k > \frac{b}{|a|} \left(-u_{\min} + \frac{u_{\min} - u_{\max}}{N} \right) \end{cases}$$

Available program: dataratelimit.m

If $a = 0$ the discrete time system evolution is

$$x_{k+1} = x_k + bhu_k$$

The applied control law (arbitrary) is

$$u_k = \begin{cases} u_{\max} & x_k \leq h \left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\max} - i \frac{u_{\max} - u_{\min}}{N - 1}, & h \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) < x_k \leq h \left(-u_{\max} + (i + 1) \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\min} & x_k > h \left(-u_{\min} + \frac{u_{\min} - u_{\max}}{N} \right) \end{cases}$$

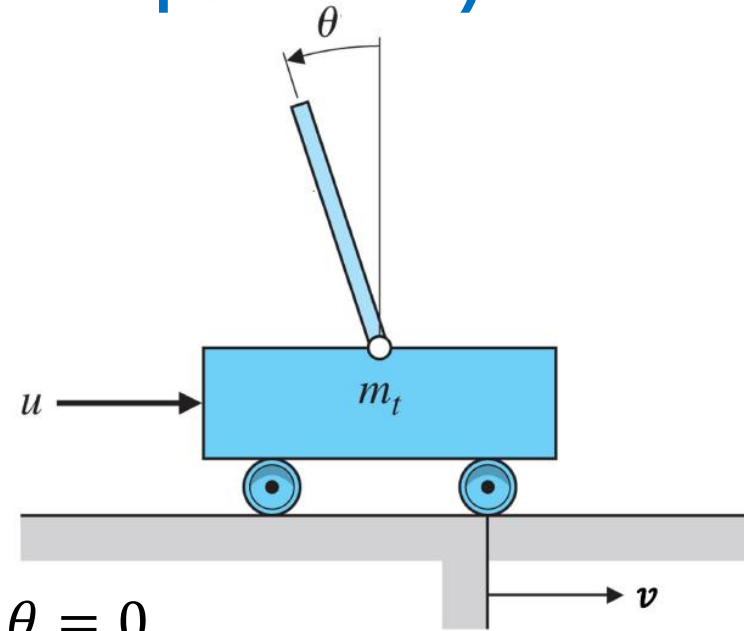
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Exercise 1: Cart-stick balancer (inverted pendulum)

Consider the cart-stick balancer system in the figure. Its model has three states, one input, and one output:

- $x_1: \theta/10$ – stick angle/10
- $x_2: \dot{\theta}/10$ – stick angular velocity/10
- $x_3 = v$ cart velocity
- u : voltage of the motor driving the wheels
- $y = 70x_1 + 30x_2 + 0.1x_3$



Control goal: maintain the stick vertical, i.e., at $\theta = 0$.

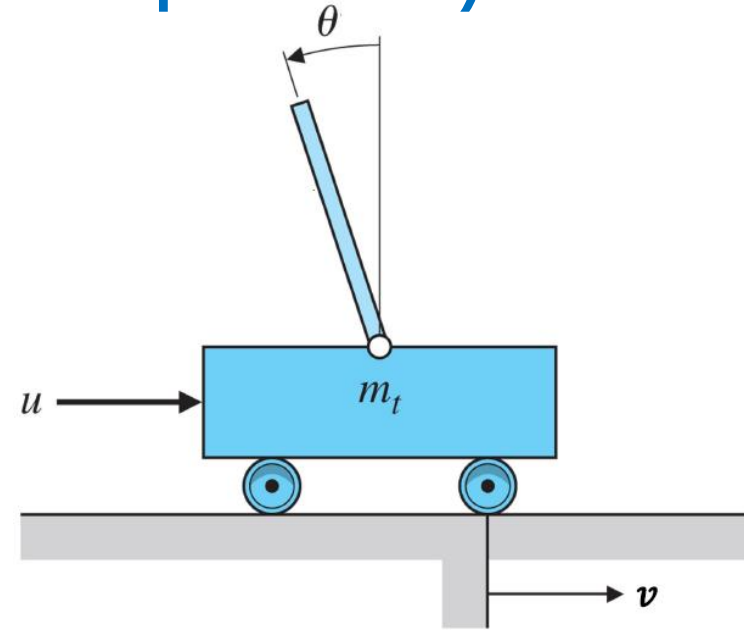
The linearized model is

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 31.33 & 0 & 0.016 \\ -31.33 & 0 & -0.216 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.649 \\ 8.649 \end{bmatrix} u$$

Assume that the system is controlled using a digital control law (i.e., with a sample-and-hold actuator). Answer the following questions considering two cases: (1) $h = 0.01$, (2) $h = 0.1$.

Exercise 1: Cart-stick balancer (inverted pendulum)

- A. Assume that the control law is of the type $u_k = Ky_k$. Compute, by trial and-error, a range of values of K that make the closed-loop system asymptotically stable and with “good” performances. Perform a number of simulations of the system trajectories starting from different initial conditions.
- B. Assume that the system is controlled through a control network, where $N_B = 1$. The control law is of the type $u_k = u_{\max} \text{sign}(y_k)$. To assess the *boundability* of the closed-loop system, perform a number of simulations of the system trajectories starting from different initial conditions. If the system displays *boundability* properties, compare the basin of attraction and the performances obtained in the following cases:
- $u_{\max} = 1$
 - $u_{\max} = 10$



Exercise 2

Consider the continuous-time system

$$\dot{x} = x + u$$

where $u_{\max} = 1$, $u_{\min} = -1$. Set the sampling time $h = \log_e(2.5) = 0.9163$ s.

- A) Setting $N_B = 1$ and using the control law $u_k = -\text{sign}(x_k)$, simulate the discrete-time evolution of the state trajectories (i.e., of x_k) starting from different random initial conditions. Show that, since $\frac{N_B}{h} < \log_2 e$, the system is not boundable.
- B) Setting $N_B = 2$ and using the control law defined in case we operate at the rate limits, simulate the discrete-time evolution of the state trajectories (i.e., of x_k) starting from different random initial conditions. Show that, since $\frac{N_B}{h} > \log_2 e$, the trajectories are bounded for some initial conditions.

Exercise 2

C) Check the performances of the control system for different values of N_B , $u_{\max}$, and $u_{\min}$. Simulate the discrete-time evolution of the state trajectories (i.e., of x_k) starting from different random initial conditions as N_B , $u_{\max}$, and $u_{\min}$ vary. Analyze the results obtained, e.g.,

- a) How do the convergence properties (convergence rate and “steady state precision”) vary as N_B increases? Do you think that, when we do not operate at the rate limit (i.e., when $\frac{N_B}{h} \gg \log_2 e$) it is possible to obtain better convergence results with a different selection function (e.g., by differently sampling the state space)?
- b) How do the trajectories vary when $u_{\max}$, and $u_{\min}$ change? For example, what happens when $u_{\min} \neq -u_{\max}$? What is the invariant set in this case? Do the trajectories “oscillate” around $x = 0$? When we do not operate at the rate limit (i.e., when $\frac{N_B}{h} \gg \log_2 e$) is it possible to devise a different selection function guaranteeing that, in “stationary conditions”, the trajectories are bounded around $x = 0$ (at least if $u_{\min} < 0$, $u_{\max} > 0$)?

Exercise 3

Consider the continuous-time system

$$\dot{x} = -x + u$$

where $u_{\max} = 1$, $u_{\min} = -1$. Set the sampling time $h = \log_e(2.5) = 0.9163$ s.

- A) Setting $N_B = 1$ and using the control law $u_k = -\text{sign}(x_k)$, simulate the discrete-time evolution of the state trajectories (i.e., of x_k) starting from different random initial conditions. Show that the trajectories are not bounded for any initial conditions.
- B) Set $N_B = 2$ and use the following control law, defined (for unstable systems) in case we operate at the rate limits:

$$u_k = \begin{cases} u_{\max} & x_k \leq \frac{b}{|a|} \left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\max} - i \frac{u_{\max} - u_{\min}}{N-1} & \frac{b}{|a|} \left(-u_{\max} + i \frac{u_{\max} - u_{\min}}{N} \right) < x_k \leq \frac{b}{|a|} \left(-u_{\max} + (i+1) \frac{u_{\max} - u_{\min}}{N} \right) \\ u_{\min} & x_k > \frac{b}{|a|} \left(-u_{\min} + \frac{u_{\min} - u_{\max}}{N} \right) \end{cases}$$

Simulate the discrete-time evolution of the state trajectories (i.e., of x_k) starting from different random initial conditions. Show that the trajectories are bounded for any initial conditions.

Exercise 3

C) Check the performances of the control system for different values of N_B , $u_{\max}$, and $u_{\min}$. Simulate the discrete-time evolution of the state trajectories (i.e., of x_k) starting from different random initial conditions as N_B , $u_{\max}$, and $u_{\min}$ vary.

Analyze the obtained results. For example, how do the convergence properties (convergence rate and “steady state precision”) vary as N_B increases? Do you think that, when we do not operate at the rate limit (i.e., when $\frac{N_B}{h} \gg \log_2 e$) it is possible to obtain better convergence results with a different selection function (e.g., by differently sampling the state space)? In particular, how it is possible to guarantee fast convergence rate and attractivity of the origin?

Exercise 4

Consider the continuous-time system

$$\dot{x} = u$$

where $u_{\max} = 1$, $u_{\min} = -1$. Set the sampling time $h = 1$ s.

- A) Setting $N_B = 1$ and using the control law $u_k = -\text{sign}(x_k)$, simulate the state trajectories starting from different random initial conditions.
- B) Setting $N_B = 2$, define a control law that guarantees that the trajectories are bounded for any initial conditions. Analyze how the system performances change (e.g., in terms of tradeoff between convergence speed and steady state “precision”) as the parameters of the selection function change.